INCH-POUND

MIL-P-52099C
12 June 1992
SUPERSEDING
MIL-P-52099B
26 August 1971

MILITARY SPECIFICATION

POSTS, HANDRAIL AND APPROACH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers handrail and approach posts used with fixed or floating bridges having aluminum deck-balk superstructures.

1.2 Classification. The posts shall be of the following types, as specified (see 6.2):

Type I - Handrail.
Type II - Approach.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

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SPECIFICATIONS

FEDERAL

PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-T-704	-	Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-889	-	Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of solicitation.

DRAWINGS

ME

TA13211E7572	-	Post, Handrail.
13211E7549	-	Post, Approach.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-JBS, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers and Welding and Brazing Operators.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

D 3953 - Strapping, Flat Steel and Seals.
D 4675 - Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

A 5.10 - Bare Aluminum and Aluminum-Alloy Welding Rods and Electrodes.
D 1.2 - Structural Welding Code - Aluminum.

(Application for copies should be addressed to the American Welding Society, 550 NW LeJeune Road, Miami, FL 33126.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION (NMFTA)

National Motor Freight Classification Rules

(Application for copies should be addressed to the National Motor Freight Traffic Association, ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The posts shall be as shown on the data list or drawing, as applicable, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are engineering design drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The item(s) shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the item may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.4 Identification marking. The post shall be metal-die stamped as shown on the applicable drawing. Die stamping shall be legible after painting.

3.5 Treatment and painting. The portions of the item normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2) top coat color shall be camouflage green, Color No. 34094. If required

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the item shall then be overcoated in accordance with the Government furnished camouflage patterns with the colors specified in the camouflage patterns.

3.6 Type I, handrail. The type I handrail post shall be as shown on TA13211E7572.

3.6.1 Welding electrodes. Welding electrodes shall conform to AWS A 5.10, classification ER4043, size as required.

3.7 Type II, approach. The type II approach post shall be as shown on drawing 13211E7549.

3.8 Government furnished property. When specified (see 6.2), the following property in the quantity indicated will be furnished by the Government (see 6.4).

Item No.	Description	Quantity
1	Camouflage paint pattern	As required

3.9 Workmanship.

3.9.1 Welders and welding.

3.9.1.1 Welders and welding operators. Before assigning any welders and welding operator to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either AWS D1.2 or the ASME code for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined. Contractors who only make horizontal welds need not qualify welders for "all position welding". The contractor is responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator (see 6.5).

3.9.1.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Aluminum welding shall be done by the shielded metal-arc or gas metal-arc (MIG) procedures in accordance with ASME or AWS Code. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for horizontal welding when practicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the

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performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.5).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.4.2, as applicable. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

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Major

101. Nonconformance to the dimensions shown on the drawings (see 3.1.1).
102. Material not as specified (see 3.3).
103. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment as specified (see 3.3.1).
104. Dissimilar metals as specified in MIL-STD-889 are not effectively insulated from each other as specified (3.3.2).
105. Contractor does not have documentation available for identification of material, material finishes, or treatments (see 3.3.3).
106. Welders and welding operators not certified as specified (see 3.9.1.1).
107. Welding and welds not as specified (see 3.9.1.2).

Minor

201. Identification marking incorrect, missing, or illegible (see 3.4).
202. Treatment and painting not as specified (see 3.5).

4.5 Inspection of packaging.

4.5.1 First article pack inspection.

4.5.1.1 Examination. The first article pack shall be examined for the defects specified in 4.5.2.3. Presence of one or more defects shall be cause for rejection.

4.5.2 Quality conformance inspection of pack.

4.5.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.5.2.3 Examination. Samples selected in accordance with 4.5.2.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

108. Boxes not in accordance with the referenced document for level A or B (see 5.1.1 and 5.1.2.
109. Boxes not reinforced as specified for level A or B (see 5.1.1 and 5.1.2).

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110. Strapping not zinc-coated for level A (see 5.1.1).
111. Gross weight of each box exceeds 200 pounds for level A or B (see 5.1.1 and 5.1.2).
112. Marking illegible, incomplete, or incorrect (see 5.2).

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. The posts of like description shall be packed in quantities not to exceed 200 pounds gross weight. The sliding lug on type I posts shall be secured with the thumbscrew. The container shall be a close fitting box conforming to PPP-B-621, class 2, style 2, for type 3 load. The ends of the box shall be reinforced by the addition of boards covering the entire inner face of each end. The reinforcing boards shall be the same thickness as the end boards, shall be placed with the grain direction at right angles to that of the end boards, and shall be nailed to them with the nails clinched. The sides, top, and bottom of the box shall be nailed to the reinforcing boards in addition to the nailing required by the box specification. The boxes shall be closed in accordance with the appendix to the box specification. Strapping shall be in accordance, with ASTM D 3953, type 1 or 2 zinc-coated, size as applicable and ASTM D 4675.

5.1.2 Level B. The posts shall be packed as specified in 5.1.1 for level A except the box shall be class 1 and strapping shall not be required to be zinc-coated.

5.1.3 Level C. The posts shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in a wooden or fiberboard container complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Type I posts are to be used as support for the handrail rope and as markers to define the roadway width of aluminum deck-balk superstructure bridges. Type II posts are used to mark the bridge approach roadways for aluminum deck-balk superstructure or other types of bridges.

6.2 Acquisition requirements. Acquisition documents should specify the following:

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a. Title, number, and date of the specification.
b. Type of post required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. Time frame required for submission of the first article model (see 3.2).
e. When camouflage pattern will be provided (see 3.5 and 6.4).
f. Level of packing required (see 5.1).
g. Any special marking (see 5.2).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.8.

6.5 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD FAR Supplement 27.475-1) exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title	Suggested Tailoring
a. 3.9.1.1	DI-MISC-80875	Welding Procedures	
b. 3.9.1.1	DI-MISC-80876	Welding Procedure Qualification Test Report	

6.6 Subject term (key word) listing.

Handrail rope
Posts

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6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - MC

Review activity:
DLA - CS

Preparing activity:
Army - ME

Project 5420-0200

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-P-52099C	2. DOCUMENT DATE (YYMMDD) 92/06/12

3. DOCUMENT TITLE
Posts, Handrail and Approach

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Violet Stewart	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3467 (2) AUTOVON 654-3467
c. ADDRESS (Include Zip Code US Army Belvoir RDE Center ATTN: STRBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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